Hyperspectral Image Intensity Adaptive Destriping Method Based on Reference Image-Guided Pixel Clustering

Tengteng Dong, Mi Wang, *Member, IEEE*, Jun Pan, *Member, IEEE*, and Qianyu Wu

***Abstract*— Hyperspectral images (HSIs) have widespread applications in geoscience, environmental monitoring, and resource management. However, in practical engineering applications, random stripe noise in HSIs severely affects data quality and accuracy, impacting the subsequent use of HSIs. In this article, we propose an HSI intensity adaptive destriping method based on reference image-guided pixel clustering. The proposed method removes stripe noise through three main stages. First, the HSI is analyzed to manually select a specific band image with minimal impact from random stripe noise. This image is designated the initial reference image, and its specific location is identified. Second, the proposed pixel threshold classification method, deep maximum interclass variance (DMIV), is used for pixel threshold classification for band images that are contaminated with random stripe noise and adjacent to the reference image. Finally, the resulting pixels are classified into two categories according to noise intensity: high intensity (HIN) and low intensity (LIN). The proposed intensity adaptive grayscale value reconstruction algorithm is used to remove stripe noise from each pixel category, and the denoised image is updated as a new reference image. Starting from the initial reference image, these steps are repeated along the spectrum to achieve destriping of all bands. We compare the proposed method against traditional and deep learning methods using real HSIs. The experimental results show that after denoising of the image using our method, both visual quality and quantitative evaluation metrics are significantly improved, with particularly excellent stripe removal performance observed for HSIs with significant grayscale variation.**

***Index Terms*— Destriping, hyperspectral images (HSIs), maximum interclass variance method, reference image, spatial clustering.**

I. Introduction

HYPERSPECTRAL remote sensing refers to the use of imaging spectrometers to capture images of the Earth's surface across a range of electromagnetic wavelengths, including visible, near-infrared, mid-infrared, and thermal infrared images. This technique simultaneously acquires images in dozens or hundreds of spectral bands and continuously records the spectral response of surface features, enabling synchronous acquisition of spatial, radiation, and spectral information from the Earth's surface. Such techniques have widespread applications in geoscience, environmental monitoring, and resource management [1], [2], [3]. However, hyperspectral images (HSIs) often contain a large amount of random stripe noise caused by the physical limitations of sensors, atmospheric effects, and calibration errors, which severely affects their subsequent application [4], [5], [6], [7]. Thus, removing random stripe noise is a necessary and important preprocessing step for application and analysis of HSIs [8], [9], [10]. In recent years, various destriping methods have been proposed, which can be roughly grouped into four categories: statistics-based [11], [12], [13], [14], filtering-based [15], [16], [17], [18], optimization-based [4], [19], [20], [21], [22], and deep learning-based [23], [24], [25], [26] methods.

Statistics-based methods assume that the digital distribution of the imaging system detectors is uniform. These methods utilize the differences in grayscale value statistics between areas polluted with stripe noise and nonpolluted areas, and the statistics of the stripe noise-polluted areas are adjusted to align with those of nonpolluted areas. The moment matching (MM) method [11], histogram matching method [13], and related improved methods [12], [14] are based on these assumptions. Statistics-based methods are straightforward and efficient for remote sensing images with relatively homogeneous land-cover types and are commonly used for destriping in current engineering applications. However, these methods often fail to effectively destripe remote sensing images with more complex land-cover types.

Filtering-based methods primarily exploit the directional properties of stripe noise in the wavelet domain and its high-frequency characteristics in the Fourier frequency domain, and thus employ appropriate filters in either the wavelet or Fourier domain to remove stripe noise. Torres et al. [7] assessed the properties of stripe noise using the scale and directional characteristics of wavelet analysis in the wavelet domain and ultimately achieved stripe noise removal using Daubechies wavelets of different orders. Chen et al. [15] proposed an adaptive wavelet filter that introduces a new parameter, the "weight sum variance of digital number probability," to adaptively select appropriate wavelet coefficients based on image characteristics for stripe noise removal. However, wavelet decomposition-based methods are influenced by

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the choice of wavelet transform function. To achieve the desired denoising results, an appropriate wavelet transform function for decomposing and processing the image must be selected. Fourier transform [17], [18] is another effective method for destriping. This method relies on the periodicity of stripe noise and works well for strongly periodic stripe noise but may fail to accurately separate stripe noise from image information in the case of nonperiodic stripe noise, potentially leading to the distortion of image information and introduction of ringing artifacts and staircase effects.

Optimization-based methods treat the destriping problem as an ill-posed inverse problem. Based on the characteristics of stripe noise, these methods construct destriping models by introducing fidelity terms, total variation (TV) models, low-rank regularization, and sparsity, thereby estimating the clean image from the noisy image. Bouali and Ladjal [19] proposed the unidirectional TV model for destriping based on analysis of the classic TV model and the directional properties of stripe noise. This model fully considers the directional information of stripe noise in the image, but fine details of the image tend to be lost during stripe noise removal. Wang et al. [20] proposed the TV-guided filtering algorithm for destriping, which aims to preserve image detail. The algorithm first employs wavelet transformation for multiscale representation of the image, then applies TV methods for edge-preserving smoothing of the image and uses the smoothed image as a guidance image, applying a guided filter in the spatial domain to remove the stripe noise. Low-rank tensor decomposition-based models [4], [21], [22] are another type of representative optimization-based destriping method. These methods decompose the noisy image into a product of multiple low-rank tensors, with each low-rank tensor representing a specific feature, such as spatial, spectral, or stripe noise components. The high-frequency characteristics of stripe noise are then used to separate and remove it, thereby improving data quality and visualization. However, optimization-based methods often have high computational cost and strong dependence on parameters, which limits their use in practical engineering applications.

Deep learning-based methods primarily use deep convolutional neural networks to learn the features of stripe noise and images in a supervised manner for destriping. The dynamical important residue network (DIRN) [23], dense net recursive convolutional neural network (DnRCNN) [24], Translution-SNet [25], and parallel high index [26] are among the advanced deep learning-based methods proposed for destriping in recent years. The performance of these methods relies heavily on the size and quality of the training dataset, which limits their generalizability. In addition, most such methods are trained using simulated stripe noise and often do not meet expectations when applied to the removal of real stripe noise. As a result, these methods are not yet effectively available for practical engineering applications.

To address HSI destriping in practical engineering applications, this article introduces an HSI intensity adaptive destriping method based on reference image-guided pixel clustering. As shown in Fig. 1, this process consists of three main stages. First, HSIs were analyzed, and an initial reference image was manually selected. Next, to perform pixel thresholding on the reference image and obtain high-precision thresholding results, we employ a deep maximum interclass variance (DMIV) method that combines deep learning concepts, the maximum interclass variance method, and a density space clustering method based on the differentiated creative search (DCS) algorithm [29] (hereafter, D/DCS). DMIV is used to classify the reference image and generate corresponding pseudo-labels. These pseudo-labels are employed for pixel threshold classification on a specific band image contaminated by random stripe noise that is adjacent to the reference image. Finally, the obtained pixel classes are categorized into two main groups, high intensity (HIN) and low intensity (LIN), based on noise intensity. The proposed intensity adaptive grayscale value reconstruction algorithm is used to remove stripe noise from each pixel class, and the denoised image is updated as a new reference image. Starting from the initial reference image, these steps are repeated along the spectrum to achieve destriping for all bands. In summary, the main contributions of this article are as follows.

1) We propose the DMIV method, which combines deep learning concepts, the maximum interclass variance method, and the D/DCS, and employ it to classify the reference image and generate corresponding pseudo-labels. These pseudo-labels are used to perform pixel threshold classification on a specific band image contaminated by random stripe noise that is adjacent to the reference image.
2) We propose a new intensity adaptive grayscale value reconstruction algorithm. The proposed method can adaptively select different strategies to remove stripe noise according to the noise intensity of different pixel categories to obtain a high-precision denoised image.
3) The experimental results show that after denoising of the image using our method, both the visual quality and quantitative evaluation metrics significantly improved; in particular, excellent stripe removal performance was observed for HSIs with significant grayscale variation.

II. Related Work

A. Maximum Interclass Variance Method

The maximum interclass variance method [27], also known as Otsu's method, assumes that the grayscale histogram of an image can be described by two Gaussian distributions, corresponding to the foreground and background, respectively. Specifically, the grayscale value of the image is considered to be composed of two pixel types: foreground pixels and background pixels. Assuming that the grayscale distribution of these two types of pixels is Gaussian, estimating the probability density of each grayscale value allows the within-class and between-class variances under each possible threshold to be calculated. Within-class variance reflects the degree of dispersion of the grayscale values of pixels in the same class. Smaller within-class variance indicates more similar grayscale values of pixels within the class. Between-class variance reflects the degree of dispersion between the foreground and background pixels. Greater between-class variance indicates

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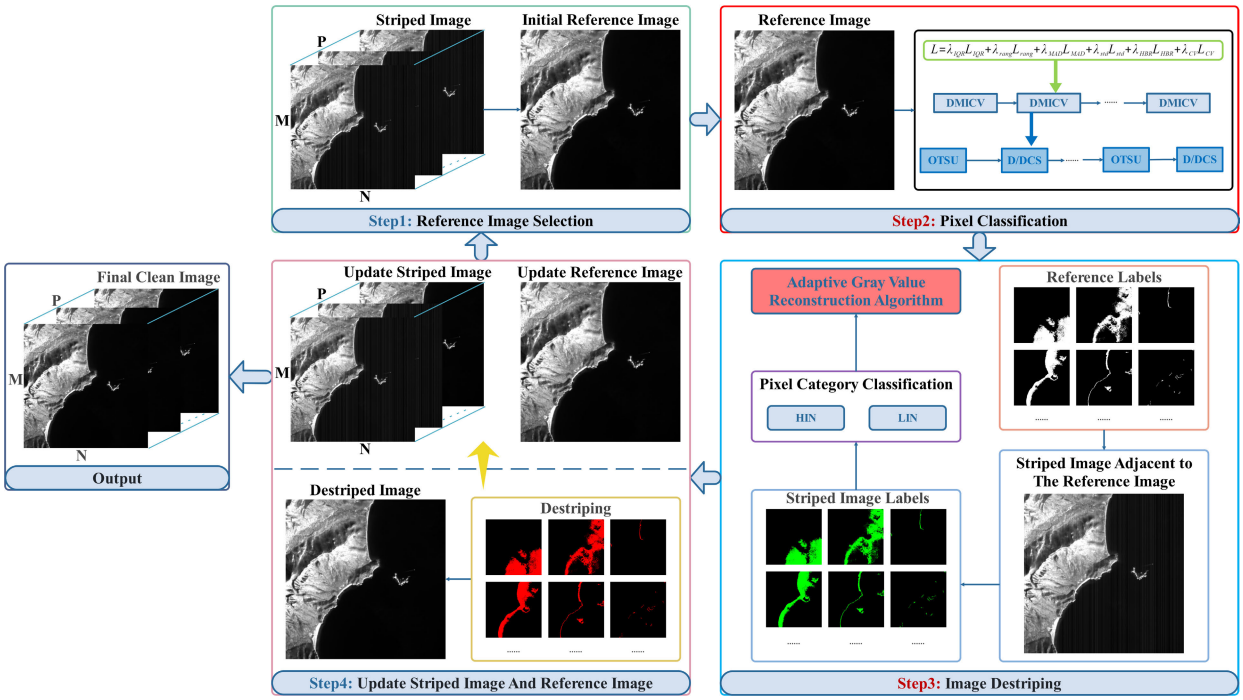


Fig. 1. Flowchart of the HSI intensity adaptive destriping method based on reference image-guided pixel clustering. First, the initial reference image is manually selected. Then the proposed DMIV-based pixel threshold classification method is used to classify reference image pixels, and corresponding pseudo-labels are generated to guide the pixel threshold classification of band images adjacent to the reference image. After the thresholding is completed, the pixel categories are divided into HIN and LIN according to noise intensity. The proposed intensity adaptive grayscale value reconstruction algorithm is used to remove stripe noise, and then the denoised image is updated as a new reference image. Starting from the initial reference image, these steps are repeated along the spectrum to achieve removal of stripe noise from all bands.

a more apparent difference between the two classes. The optimal threshold is determined by minimizing within-class variance or maximizing between-class variance, and thus is optimal in the least squares sense. The specific steps of this method are as follows. Let the image size be $M \times N$ and the optimal segmentation threshold be T. The number of pixels belonging to the background is N_0, while the number of pixels belonging to the foreground is N_1. The proportion of background pixels in the entire image is ω_0, with a mean grayscale value of μ_0. The proportion of foreground pixels is ω_1, with a mean grayscale value of μ_1. The mean grayscale value of the entire image is μ. Therefore, $\omega_0 = N_0/(M \times N)$, $\omega_1 = N_1/(M \times N)$, and $N_0 + N_1 = M \times N$. And thus, $\omega_0 + \omega_1 = 1$ and $\mu = (\mu_0 * N_0 + \mu_1 * N_1)/(M \times N)$.

In this method, the within-class variance $\sigma_W^2 = \omega_0\sigma_0^2 + \omega_1\sigma_1^2$ and between-class variance $\sigma_B^2 = \omega_0(\mu_0 - \mu)^2 + \omega_1(\mu_1 - \mu)^2 = \omega_0\omega_1(\mu_0 - \mu_1)^2$ are defined, where σ_0^2 and σ_1^2 represent the variances of the background and foreground pixels, respectively. Pixels of the same class ideally have very small within-class variance; thus, greater between-class variance between the background and foreground indicates a larger difference between the two parts of the image. The optimal value is sought for the segmentation threshold T that minimizes within-class variance or maximizes between-class variance and is obtained using the traversal method.

However, the use of Otsu thresholding alone for pixel thresholding has a notable drawback: this method can only divide pixels into two classes, and its thresholding accuracy is often low, which means that each class frequently contains pixels that do not belong to it. Therefore, in this study, we employ a cyclic iterative processing approach, repeatedly applying the proposed DMIV to classify pixels of the target image until all pixel categories are obtained. This approach addresses the shortcomings of Otsu thresholding, which can only classify pixels into two categories and has lower thresholding accuracy.

B. Density-Based Spatial Clustering of Applications With Noise

The traditional density-based spatial clustering of applications with noise (DBSCAN) algorithm [28] characterizes data density using two parameters: the maximum neighborhood radius ϵ and the minimum number of points MinPts. Under the density-based definition, DBSCAN classifies data points into three types: core points, which are points with at least MinPts points within their ϵ-neighborhood; border points, which are not core points but lie within the ϵ-neighborhood of a core point; and noise points, which are neither core points nor border points. For each core point, all points within its ϵ-neighborhood (including other core points) are interconnected to form a cluster, and border points are assigned to the cluster of the nearest core point. Finally, points that are not assigned to any cluster are labeled as noise. Due to the clustering principle of DBSCAN, selecting appropriate values for the maximum neighborhood radius ϵ and minimum points MinPts based on the characteristics of the dataset is crucial to effective clustering. Therefore, in this study, we perform parameter optimization and performance

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enhancement on DBSCAN using the recently introduced DCS algorithm to determine the most appropriate maximum neighborhood radius ϵ and minimum number of points MinPts and thus achieve more efficient data clustering.

C. DCS Algorithm

The DCS algorithm, proposed by Duankhan et al. [29], represents a novel intelligent optimization approach inspired by differentiated knowledge acquisition and creative realism. This algorithm exhibits strong evolutionary capabilities, fast search speed, and outstanding optimization capability, enabling efficient application to various prediction and optimization problems for complex scenarios. Therefore, in this study, we used DCS for parameter optimization and performance enhancement of DBSCAN, proposing a D/DCS. Here, we briefly describe the implementation of DCS. The implementation process is as follows.

1) *Initialization:* In DCS, the optimization process begins with a set of candidate solutions, which are randomly generated between the upper bound (UB) and lower bound (LB) of the problem to be optimized. The specific generation method is detailed in [29].
2) *Differentiated Knowledge Acquisition:* The foundation of differentiated knowledge acquisition lies in the variable potential for knowledge acquisition among humans, with a focus on the speed of acquiring new knowledge, and its impact on the parameters of candidate solutions is multifaceted. This method primarily evaluates the adaptability of each candidate solution based on the problem's objective function or evaluation criteria by assigning superior adaptability parameters to better-performing solutions and promoting diversity among solutions with similar adaptability levels to achieve differentiated knowledge application.
3) *Creative Realism:* In the DCS algorithm, innovative elements are integrated into the adaptability adjustment mechanism to enhance the algorithm's exploration capacity and problem-solving creativity. DCS focuses on modeling of the problem to be solved, including both systematic optimization methods and a combination of innovative exploration strategies.
4) *Iterative Updating of Candidate Solutions:* DCS primarily uses the acquired differentiated knowledge to adaptively adjust candidate solutions during the solving process, replacing poorly performing candidate solutions with new ones.
5) *Retrospective Assessment:* This is a critical step in DCS. Retrospective assessment (RA) achieves the establishment of evaluation criteria for assessing the effectiveness and efficiency of the algorithm through the following three processes: assessing the quality of solutions, comparing the effectiveness of different strategies, and assessing algorithm convergence. First, its helps the algorithm to evaluate the quality of the generated solutions, specifically whether these solutions meet the requirements and constraints of the optimization problem. Second, the DCS algorithm uses various knowledge acquisition methods and creative realism strategies. RA helps to compare the performance of these strategies at solving problems, identifying the strategies that are best suited for specific types of optimization problems. Third, RA helps to identify performance trends and potential improvement points by analyzing whether the solutions progressively converge on an optimal solution. The specific assessment methods have been detailed previously [29].

D. Standard MM and Improved Algorithms

The standard MM method [11] is a statistical approach for destriping. This method normalizes the mean and standard deviation of the response data from each sensor based on linear relationships, adjusting them to a common reference value to effectively remove stripe noise. The standard MM method assumes that the charge-coupled device (CCD) sensor has a linear response function within the ground spectral response range and exhibits translation invariance. Let C_i be the ith CCD pixel, and its spectral response function can be expressed as follows:

$$Y_i = k_i X + b_i \tag{1}$$

where Y_i is the grayscale value of the image, which is the spectral response value of the detector C_i. X is the radiation component value received by the pixel, and k_i and b_i are the gain value and offset value, respectively, for pixel i.

Based on the spectral response function, if each CCD pixel receives the same radiation intensity X and the image scene is evenly distributed, different CCD pixels will have different gain values k_i and offset values b_i, resulting in different pixel grayscale values Y_i. If the image scene is evenly distributed, different Y_i values appear as stripe noise. If Y_i can be normalized to a common value Y, the stripe noise can be effectively removed.

Assuming that the image scene is evenly distributed, when CCDs scan and acquire ground image data by column, the mean and variance of the radiation intensity for the same pixel are approximately equal across columns. Therefore, stripe noise is removed by adjusting the mean and standard deviation of each sensor to those reference values based on the entire image. This is the concept of MM method, expressed as follows:

$$Y = \frac{\sigma_r}{\sigma_i} X + \mu_r - \mu_i \frac{\sigma_r}{\sigma_i} \tag{2}$$

where X and Y denote the image grayscale values of each pixel in the ith column before and after correction, respectively. μ_r and σ_r represent the mean and standard deviation of grayscale values for the entire image, while μ_i and σ_i denote the mean and standard deviation of grayscale values for the ith column of the image.

The standard MM method uses the mean and standard deviation of pixel grayscale values across the entire image as the reference mean and standard deviation, and all column means and standard deviations are adjusted to this reference. However, the assumptions about data distribution limit the effectiveness of stripe noise removal. Therefore, when the

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distribution of ground objects is complex, the expected effect cannot be achieved. To solve this problem, many scholars have proposed improved methods based on standard MM. Improved MM methods based on mean filtering [31] and polynomial fitting [32] show relatively strong improvements among MM methods.

The basic idea of the improved MM method based on mean filtering is as follows. First, the column means and variances of the original image are calculated. Then these column means and variances are segmented and averaged. Next, these averaged column means and variances are used to fit the smoothed filtered column means and variances. Substituting these mean and variance values into the standard MM equation produces grayscale values of the image after destriping. Another effective improvement method is the polynomial fitting-based MM method. Fundamentally, this method involves initially calculating the column means and variances of the image containing stripe noise. Then the least squares method is used to fit the column means and variances of the original image, obtaining a set of fitting coefficients. Subsequently, a polynomial is generated based on the fitting coefficients and used to compute reference means and variances for each column. Finally, the computed reference means and reference variances are used to remove the stripe noise.

Although the above improved MM method has produced good results, significant limitations remain because its performance strongly depends on the statistical distribution of data for each sensor. Therefore, in this study, we first perform pixel thresholding classification on the noisy image to ensure that each pixel class meets the assumption that the image scene is evenly distributed. Next, by combining the standard MM method with the existing improved MM methods, we propose an intensity adaptive grayscale value reconstruction algorithm that adaptively selects different methods for removal of stripe noise based on the noise intensity of each pixel class.

III. Methodology

To address the problem of random stripe noise in HSIs and enhance the universality of the proposed stripe noise removal method for engineering applications, we propose an HSI intensity adaptive destriping method based on reference image-guided pixel clustering. This method removes stripe noise through three main stages: selection of the initial reference image, pixel threshold classification based on DMIV guided by the reference image, and intensity adaptive stripe noise removal. The overall process is illustrated in Fig. 1.

When selecting the initial reference image, we analyzed in detail the sensitivity of each band of the HSI to random stripe noise, manually selected the band image that was least affected by stripe noise as the initial reference image, and located its specific position.

For the pixel threshold classification method proposed in this study, we adopt the idea of deep learning, proposed a new loss function, and combine it with the maximum interclass variance method and D/DCS to obtain a new pixel threshold classification method, designated the DMIV method. At the same time, to perform threshold classification for all pixels of the image, we use a loop iterative processing method to repeatedly apply the proposed DMIV to classification of reference image pixels until all pixel categories are obtained. We generate corresponding pseudo-labels based on the thresholding results to guide the pixel threshold classification of specific band images adjacent to the reference image that are contaminated by random stripe noise.

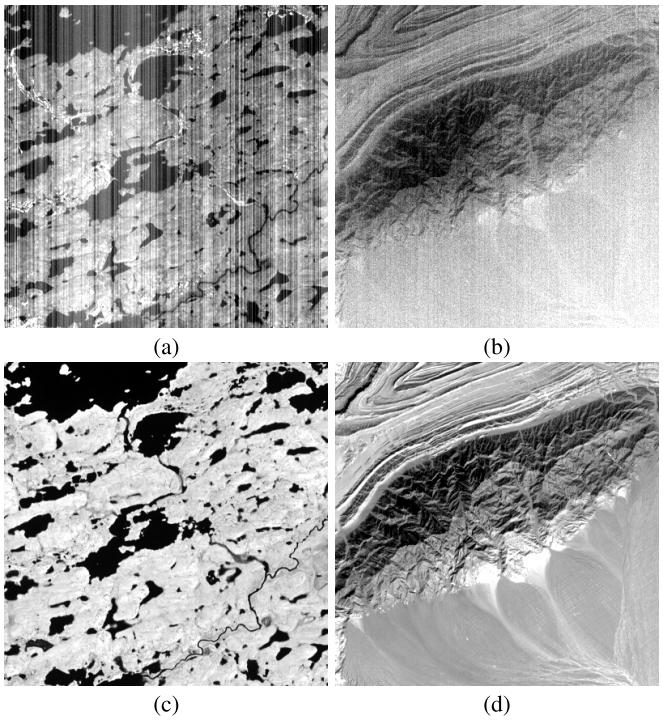


Fig. 2. Visible light and near-infrared images designated ZY1F and GF5B. (a) ZY1F visible light image. (b) GF5B visible light image. (c) ZY1F near infrared image. (d) GF5B near infrared image.

After completing pixel threshold classification, we divide the obtained pixel classes into two categories according to noise intensity: HIN and LIN. Using the proposed intensity adaptive grayscale value reconstruction algorithm, we adaptively remove the stripe noise of different pixel categories according to the noise intensity and use the denoised image as a new reference image. Starting from the initial reference image, we repeat the above steps along the spectrum until the removal of stripe noise from all bands is achieved. Below, we explain each link of the denoising process in detail.

A. Selection of the Initial Reference Image

In this study, we analyze in detail the sensitivity of each HSI band to noise. The sensitivity of bands to noise is usually related to their wavelength [36]. Generally speaking, short wavelengths such as purple light and visible light are strongly affected by atmospheric scattering, particularly Rayleigh scattering, which is particularly significant in the ultraviolet and blue light regions. When the receiving sensitivity of the sensor is low, the short-wavelength region is most susceptible to the impact of random band noise. In the mid-wavelength region, such as the red region of visible light and the near-infrared region, bands usually have higher signal-to-noise ratios on ground objects with higher reflectivity (such as vegetation),

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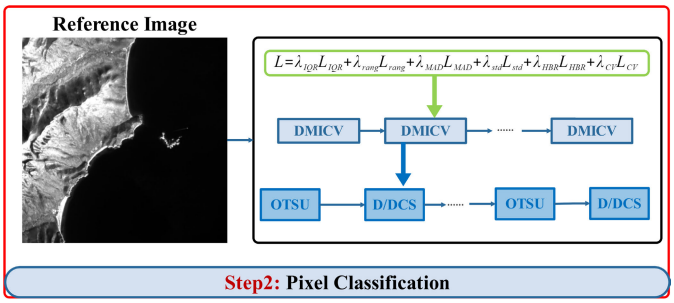


Fig. 3. DMIV structure diagram. DMIV is composed of the loss function L, Otsu thresholding, and the D/DCS.

as they are in the atmospheric window (the wavelength region for which atmospheric absorption is weak), and thus are less affected by random stripe noise. For long wavelengths, such as the short-wave infrared and long-wave infrared regions, bands are usually less sensitive to noise due to the strong effects of thermal radiation and relatively minor atmospheric absorption and scattering effects, which make them less susceptible to random stripe noise. Therefore, in this study, we generally select the band image with the least influence from stripe noise in medium- and long-wavelength regions through visual interpretation as the initial reference image and locate its specific position.

Taking the ZY1F and GF5B visible to near-infrared (VN) HSIs obtained from the advanced hyperspectral imager used in this study as examples, we analyzed and visually interpreted their various band images and found that band images in the short-wavelength region (mainly the visible light region) were apparently affected by random stripe noise to varying degrees, as shown; band images in the medium-wavelength region (mainly the near-infrared region) were little affected by random stripe noise (Fig. 2). Therefore, in this study, we selected the 48th band image of the ZY1F VN HSI as the reference image, with a wavelength of 790.3780 nm, using the 94th band image of the GF5B VN HSI as the reference image, with a wavelength of 784.7032 nm.

B. Pixel Threshold Classification Based on DMIV

After selecting the initial reference image, we use an iterative processing method in a loop to repeatedly apply the proposed DMIV to classification of reference image pixels until all pixel categories are obtained, thereby generating corresponding pseudo-labels to guide the pixel classification of other band images. DMIV is a pixel threshold classification module combining deep learning ideas, Otsu thresholding, and our proposed D/DCS. This method mainly consists of the loss function L, Otsu thresholding, and D/DCS. The specific structure is illustrated in Fig. 3.

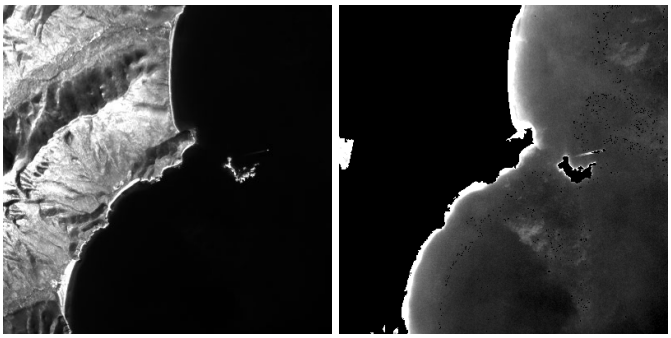
1) D/DCS: Density Space Clustering Method Based on the DCS Algorithm: To address the problem of low Otsu thresholding accuracy, we use the proposed D/DCS to remove abnormal pixels from the Otsu thresholding results. The traditional DBSCAN algorithm [28] mainly uses two parameters, the maximum community radius ϵ and the minimum number of points MinPts, to characterize the density of the data and remove noise. Thus, the selection of reasonable values for ϵ and MinPts based on the characteristics of the dataset plays a decisive role in the clustering efficacy of DBSCAN. Therefore, in this study, we use the DCS algorithm to optimize the parameters of DBSCAN, obtaining the most reasonable ϵ and MinPts values possible to achieve efficient data clustering analysis. A D/DCS is proposed to remove abnormal pixels from the Otsu threshold classification results. D/DCS is described in detail below.

When using D/DCS to remove abnormal pixels from the classification results, the process can be divided into the following main steps.

1) *Construction of the Fitness Function:* In this study, we employ the silhouette coefficient of the clustering result as the fitness function. The silhouette coefficient is an indicator used to evaluate the clustering effect, which describes the clarity of the outline of each category after clustering. This indicator contains two factors, compactness and separation. Compactness reflects the closeness between a sample point and elements within the class, while separation reflects the closeness between a sample point and elements outside the class. The value range of the silhouette coefficient is $[-1, 1]$. A value closer to 1 indicates better clustering performance; conversely, a value closer to -1 indicates worse clustering performance.
The silhouette coefficient of a single sample i is defined as follows:
$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{3}$$
where a_i represents the average distance between sample i and other samples in cluster A, as described in (4). If only one sample i is present in the cluster, then $s(i) = 0$
$$a(i) = \text{average}_{j \in A, j \neq i}(\text{dist}(i, j)) \tag{4}$$
where b_i represents the minimum value of the average distance between sample i and samples in other cluster B, as expressed in the following equation:
$$\begin{cases} \text{dist}(i, B) = \text{average}_{j \in B, A \neq B}(\text{dist}(i, j)) \\ b(i) = \min_{A \neq B} \text{dist}(i, B). \end{cases} \tag{5}$$
From this equation, we can derive the silhouette coefficient of the clustering set, which is the average of the silhouette coefficients of all samples, as described in the following equation:
$$S = \frac{1}{N} \sum_{i=1}^{N} s(i) \tag{6}$$
where N is the total number of samples.

2) *DCS Optimization:* Duankhan et al. [29] provided a detailed description of the DCS algorithm and its solution process. In this study, we directly employ the DCS algorithm to optimize the constructed fitness function and thus maximize the silhouette coefficient. In this manner, the most reasonable values for ϵ and MinPts are obtained for the removal of abnormal pixels.

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(a) (b)

Fig. 4. OTSU threshold classification result. (a) Original image. (b) OTSU threshold classification result.

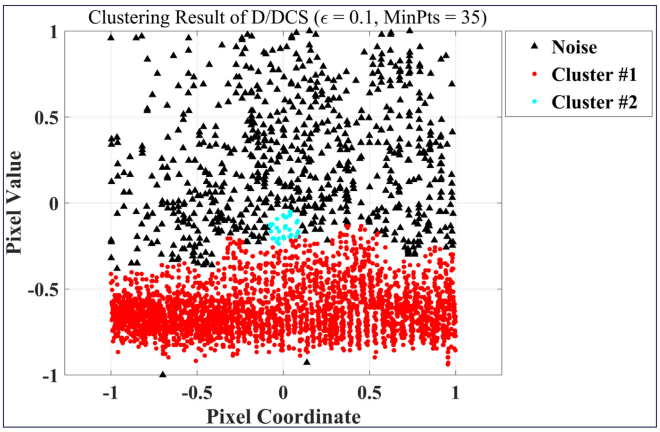


Fig. 5. D/DCS clustering result map.

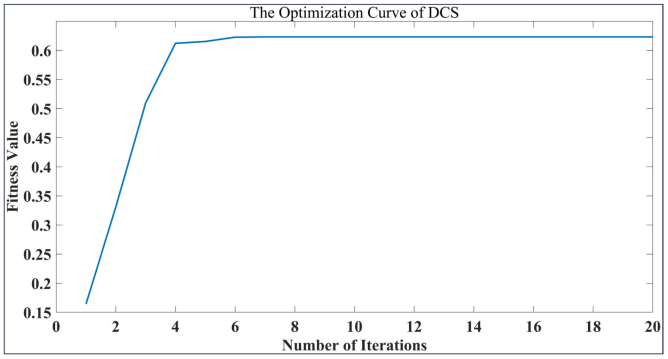


Fig. 6. Optimization curve map when using the DCS algorithm to determine the most reasonable maximum community radius ϵ and the minimum number of points MinPts.

3) *D/DCS Clustering:* The values of ϵ and MinPts obtained in 2) are used to perform cluster analysis on the pixel set to identify the cluster structure in the pixel set for processing of abnormal pixel points. After mapping of the Otsu threshold classification results shown in Fig. 4 to values between -1 and 1, D/DCS is used to cluster the normalized threshold classification results presented in Fig. 4, and the results are shown in Fig. 5. Fig. 6 presents the DCS optimization curve obtained. Figs. 5 and 6 show that D/DCS has enhanced clustering accuracy and stability, and thus better reflects the internal structure and rules of the pixel set, providing reliable support for the subsequent removal of abnormal pixel values.

2) Pixel Threshold Classification: Here, to obtain multiple high-precision pixel threshold classification results without abnormal pixels, we propose a cyclic iterative processing method. In each iteration, D/DCS is used to process the pixel threshold classification results of the Otsu method, and noise in the clustering results as well as the category with the smallest number of pixels are removed. At the same time, a loss function is proposed to determine whether the optimal processing result is obtained, as follows:

$$L = \lambda_{\mathrm{IQR}} L_{\mathrm{IQR}} + \lambda_{\mathrm{rang}} L_{\mathrm{rang}} + \lambda_{\mathrm{MAD}} L_{\mathrm{MAD}} + \lambda_{\mathrm{std}} L_{\mathrm{std}} + \lambda_{\mathrm{HBR}} L_{\mathrm{HBR}} + \lambda_{\mathrm{CV}} L_{\mathrm{CV}} \quad (7)$$

where L_{IQR}, L_{rang}, L_{MAD}, L_{std}, L_{HBR}, and L_{CV} represent the interquartile range, range, mean difference, standard deviation, variation ratio, and coefficient of variation in the pixel threshold classification results, respectively. These indicators are all used to describe the degree of data dispersion, and their calculation methods are as follows. For these indicators, a smaller absolute value represents more concentrated data. λ_{IQR}, λ_{rang}, λ_{MAD}, λ_{std}, λ_{HBR}, and λ_{CV} are the weight coefficients corresponding to those indicators, and they are of the same size

$$L_{\mathrm{IQR}} = Q_3 - Q_1 \quad (8)$$

$$L_{\mathrm{rang}} = X_{\max} - X_{\min} \quad (9)$$

$$L_{\mathrm{MAD}} = \frac{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)}{n} \quad (10)$$

$$L_{\mathrm{std}} = \sqrt{\frac{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2}{n}} \quad (11)$$

$$L_{\mathrm{HBR}} = \frac{n - f_m}{n} \quad (12)$$

$$L_{\mathrm{CV}} = \sqrt{\frac{\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2}{n}} / \bar{X} \quad (13)$$

where Q_1, Q_2, and Q_3 are the three points that divide all data into four equal parts after sorting of a group of data of length n. The values corresponding to the positions of these three points are called quartiles, which are recorded as Q_1 (first quartile), Q_2 (second quartile), and Q_3 (third quartile). X_i is an element in the array, $\bar{X}$ is the mean of the array, and f_m is the mode of the array.

When the loss function tends to be stable, we consider the abnormal pixels in the pixel threshold classification results to have been completely removed, and the Otsu segmentation threshold obtained at this time is the optimal segmentation threshold. Based on the optimal segmentation threshold obtained, we can separate a class of pixels from the remote sensing image and remove the separated pixel class from the original image. Then, we continue to classify the original image with the pixel class removed until all pixels in the original image are completely classified, thereby obtaining multiple pixel threshold classification results that contain no abnormal pixels. The specific process is presented in Algorithm 1.

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Algorithm 1 DMIV Threshold Classification Algorithm

Input: Clean sub-image Y
Output: Pixel classification set $O = [O_1, O_2, \ldots, O_m]$
Initialize: $i = 0, Y^0 = Y$
  While $Y^i \neq \emptyset$
    $T^i = OTSU(Y^i)$
    $O^i = Y^i(Y^i \leq T^i)$
    Initialize: $j = 0, O_j = O^i$
    While $True$
      $Temp = D/DCS(O_j)$
      $T^j = OTSU(Temp)$
      $O_{j+1} = Temp(Temp \leq T^j)$
      $L^j = L(O_{j+1})$
      $j++$
    If L^j **stabilizes the loop ends**
    $O.push_back(O_{j+1})$
    $Y^{i+1} = Y^i - O_{j+1}$
    $i++$
  end

C. Intensity Adaptive Destriping

After threshold classification using DMIV is completed, we divide the pixels into two categories according to noise intensity, HIN and LIN, and combine the standard MM method with the existing improved MM method to adaptively select methods for stripe noise removal from pixel categories according to noise intensity. The pixel categories are merged after denoising to obtain the final denoised image.

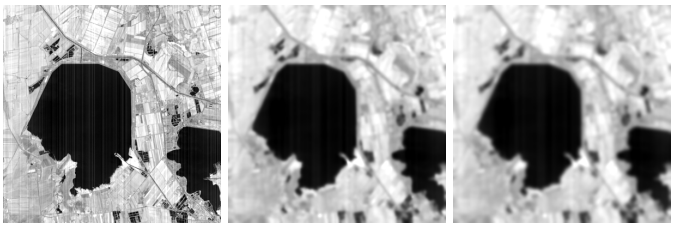
1) Stripe Noise Intensity Estimation: To more accurately estimate the noise intensity of pixel categories, we propose a new stripe noise intensity estimation method. First, the original noisy image Y is filtered multiple times using bilateral filters (BFs) to obtain a smooth image $\hat{Y}$ without stripe noise. Then the following equation is used to estimate the stripe noise S, where $\hat{S}, Y, \hat{Y} \in R^{m \times n}$. m and n are the numbers of rows and columns in the image, respectively,

$$\hat{S} = Y - \hat{Y}. \tag{14}$$

BF is used to perform multiple filtering processes on the original noisy image because compared to other smoothing filters, it can better protect image details while completely removing stripe noise [30], thereby allowing for more accurate estimation of the stripe noise. In Fig. 7, we compare the smoothing results of BF and the Gaussian filter. Compared to the Gaussian filter, the details of the smoothing results of BF are more apparent.

After estimating the stripe noise $\hat{S}$, we use the pseudo-labels generated as described above to classify $\hat{S}$ into categories $[\hat{S}_1, \hat{S}_2, \ldots, \hat{S}_i]$, and then use the standard deviation to represent the intensity $I_{\hat{S}_i}$ of the categories of stripe noise, as described by the following equation:

$$I_{\hat{S}_i} = \sqrt{\frac{\sum_{k=1}^{N} \left(\hat{S}_i^k - \langle \hat{S}_i \rangle\right)^2}{N}} \tag{15}$$



(a) (b) (c)

Fig. 7. Comparison of Gaussian filtering and BF filtering smoothing results. (a) Noise image. (b) BF filtering. (c) Gaussian filtering.

where $\hat{S}_i^k$ is the kth pixel value of the ith category, $\langle \hat{S}_i \rangle$ is the average pixel value of the ith category, and N is the number of pixels in the ith category.

2) Intensity Adaptive Grayscale Value Reconstruction Algorithm: In engineering applications, statistics-based methods are currently the most widely used stripe noise removal approach. Commonly used methods are the MM method and the histogram matching method. The denoising effects of these two methods are comparable, but the MM method has higher computational efficiency and is more suitable for removing stripe noise from remote sensing images with relatively homogeneous land-cover types. Therefore, we use the MM method to remove stripe noise from multiple pixel categories. However, different MM methods show differential performances in terms of removing stripe noise of different intensities. In Section IV-E, we discuss in detail the performance of the standard MM method [11], the improved MM method based on mean filtering [31], and the improved MM method based on polynomial fitting [32]. For stripe noise of all tested intensities, the denoising effect of the improved MM method based on mean filtering is worst. The standard MM method and the improved MM method based on polynomial fitting exhibit different denoising performances as noise intensity increases; when the noise intensity is high, the standard MM method performs better, while at low noise intensity, the improved MM method based on polynomial fitting performs better.

Here, we propose an intensity adaptive grayscale value reconstruction algorithm for removal of stripe noise from multiple pixel categories through adaptive selection of methods according to noise intensity

$$X_i(m,n) = \alpha_i \times \mathrm{MM}_i(m,n) + (1-\alpha_i) \times \mathrm{PFMM}_i(m,n) \tag{16}$$

where $X_i(m,n)$, $\mathrm{MM}_i(m,n)$, and $\mathrm{PFMM}_i(m,n)$ are the stripe noise removal result for the ith pixel category, the denoising result of the standard MM method, and the denoising result of the improved MM method based on polynomial fitting, respectively; m and n represent the number of rows and columns of the image, respectively; and α_i is calculated as follows:

$$\alpha_i = \frac{I_{\hat{S}_i}}{\mathrm{std}(Y_i) + I_{\hat{S}_i}} + 0.5 \tag{17}$$

where $\mathrm{std}(Y_i)$ is the standard deviation among pixel categories.

When α_i is large, i.e., when the intensity of stripe noise in the ith pixel category is high, the standard MM method

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tends to be used; when it is small, i.e., when the intensity of stripe noise in the ith pixel category is low, the improved MM method based on polynomial fitting tends to be used. Equation (16) is used to process all pixel categories $[Y_1, Y_2, \ldots, Y_i]$ separately and obtain denoising results $[X_1, X_2, \ldots, X_i]$ for each pixel category. Then, (18) is employed to merge all pixel categories and thus complete the removal of stripe noise

$$X = \sum_{i=1}^{n} X_i \tag{18}$$

where X is the final denoising result.

IV. Experimental Results and Analysis

In this section, we use real HSIs to experimentally analyze the stripe noise removal capacity, effective information retention capacity, and operating efficiency of the proposed method. To verify the superiority of the proposed method, we compare it with six state-of-the-art HSI stripe noise removal methods: MM [11], DIRN [23], DnRCNN [24], double low-rank (DLR) [33], anisotropic spatial–spectral TV regularized DLR (ATVDLR) [34], and spatial–spectral TV based on the $\ell_{2,1}$ nonconvex metric ($\ell_{2,1}$-SSTV) methods [35]. In addition, we conduct in-depth analysis of the performance of the pixel threshold classification method proposed in this study, the standard MM algorithm, and the improved MM algorithm.

A. Experimental Data and Evaluation Metrics

We used *L*1-A data for the ZY1F HSI and GF5B HSI to perform random stripe noise removal experiments. The spatial resolution of ZY1F and GF5B is 30 m, and some bands of these two images are affected by random stripe noise. After selection of the reference image, four subimages of size $400 \times 400 \times 45$ and $400 \times 400 \times 35$ are cropped from the ZY1F and GF5B HSIs, respectively, and mapped to a grayscale range of [0, 1] for comprehensive verification and analysis of the stripe removal performance of the proposed method.

The stripe noise removal capacity, information retention capacity, and operating efficiency of the proposed method are quantitatively evaluated using the mean noise removal rate (MNR), mean peak signal-to-noise ratio (MPSNR), mean structural similarity index (MSSIM), mean spectral angle distance (MSAD), and denoising time (T). Higher MNR and MPSNR values indicate better stripe noise removal capacity; higher MSSIM values and lower MSAD values indicate that effective information of the denoised image is better retained; and smaller T represents higher operating efficiency of stripe noise removal. The methods for calculating MNR, MPSNR, MSSIM, and MSAD are as follows:

$$\text{MNR} = \frac{1}{d}\sum_{k=1}^{d}\frac{N_0^k}{N_1^k} \tag{19}$$

$$\text{MPSNR} = \frac{1}{d}\sum_{k=1}^{d} 20\log_{10}\frac{1}{\sqrt{\text{MSE}}}$$

$$\text{s.t. MSE} = \frac{1}{MN}\sum_{i=0}^{M-1}\sum_{j=0}^{N-1}\left[X_k(i,j) - \hat{X}_k(i,j)\right]^2 \tag{20}$$

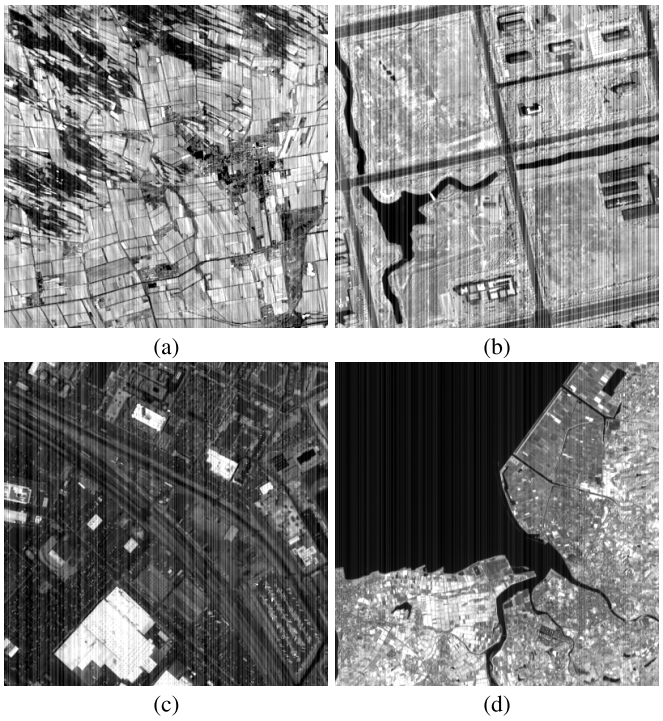


Fig. 8. Real hyperspectral noise image. (a) and (b) Test1–Test2: GF5B HIS. (c) and (d) Test3–Test4: ZY1F HSI.

$$\text{MSSIM} = \frac{1}{d}\sum_{k=1}^{d}\frac{(2\mu_X\cdot\mu_{\hat{X}}+c_1)(2\sigma_{X\hat{X}}+c_2)}{\left(\mu_X^2+\mu_{\hat{X}}^2+c_1\right)\left(\sigma_X^2+\sigma_{\hat{X}}^2+c_2\right)} \tag{21}$$

$$\text{MSAD} = \frac{1}{MN}\sum_{i=0}^{M-1}\sum_{j=0}^{N-1}\arccos\left(\frac{X_{ij}\cdot\hat{X}_{ij}}{\left\|X_{ij}\right\| * \left\|\hat{X}_{ij}\right\|}\right) \tag{22}$$

where N_0^k represents the power of the frequency component of the kth band noise image; N_1^k represents the power of the frequency component of the kth band destriped image; d is the number of bands; M and N are the number of rows and columns in the image, respectively; X and $\hat{X}$ are the destriped image and the reference image, respectively; μ_X and $\mu_{\hat{X}}$ are the means of X and $\hat{X}$, respectively; $\sigma_{X\hat{X}}$ is the covariance of X and $\hat{X}$; σ_X and $\sigma_{\hat{X}}$ are the standard deviations of X and $\hat{X}$, respectively; c_1 and c_2 are constants greater than 0; and $\|\cdot\|$ represents the square root of the sum of squares of each number in the matrix.

B. Evaluation of Experimental Results

To objectively and comprehensively validate the stripe noise removal capacity, effective information retention capacity, and operating efficiency of the proposed method, we conduct stripe noise removal experiments using four real HSIs with significantly different levels of stripe noise interference, complexities of ground object distribution, and amplitudes of grayscale variation, as shown in Fig. 8. The selected experimental data represent different types of stripe noise-polluted images.

We employ our method and the comparative method to remove stripe noise from the real HSIs shown in Fig. 8. The removal results are shown in Figs. 9–12, with red boxes highlighting enlarged views of the original noisy images and denoised images. Next, we conduct detailed analysis

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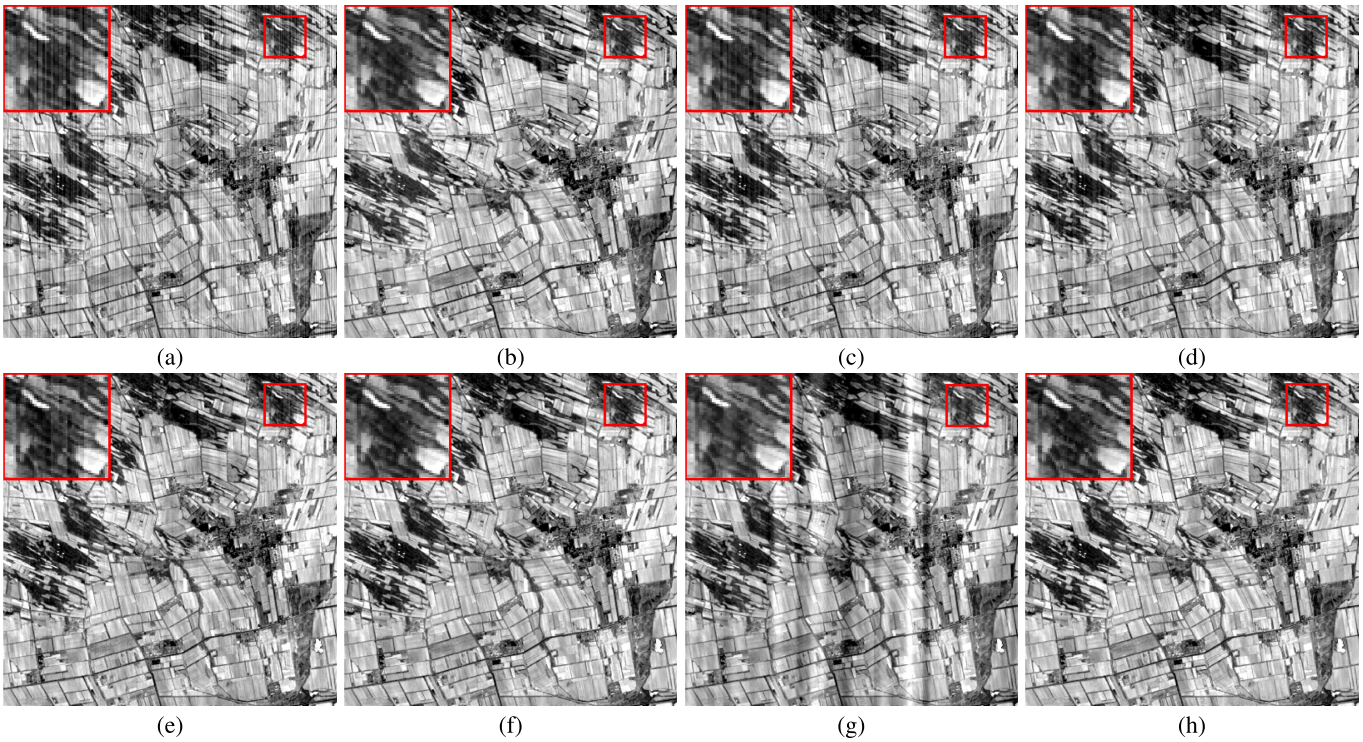


Fig. 9. Corrected Test1 for band 4. (a) Original noise image. (b) MM. (c) DIRN. (d) DnRCNN. (e) DLR. (f) ATVDLR. (g) $\ell_{2,1}$-SSTV. (h) Our method.

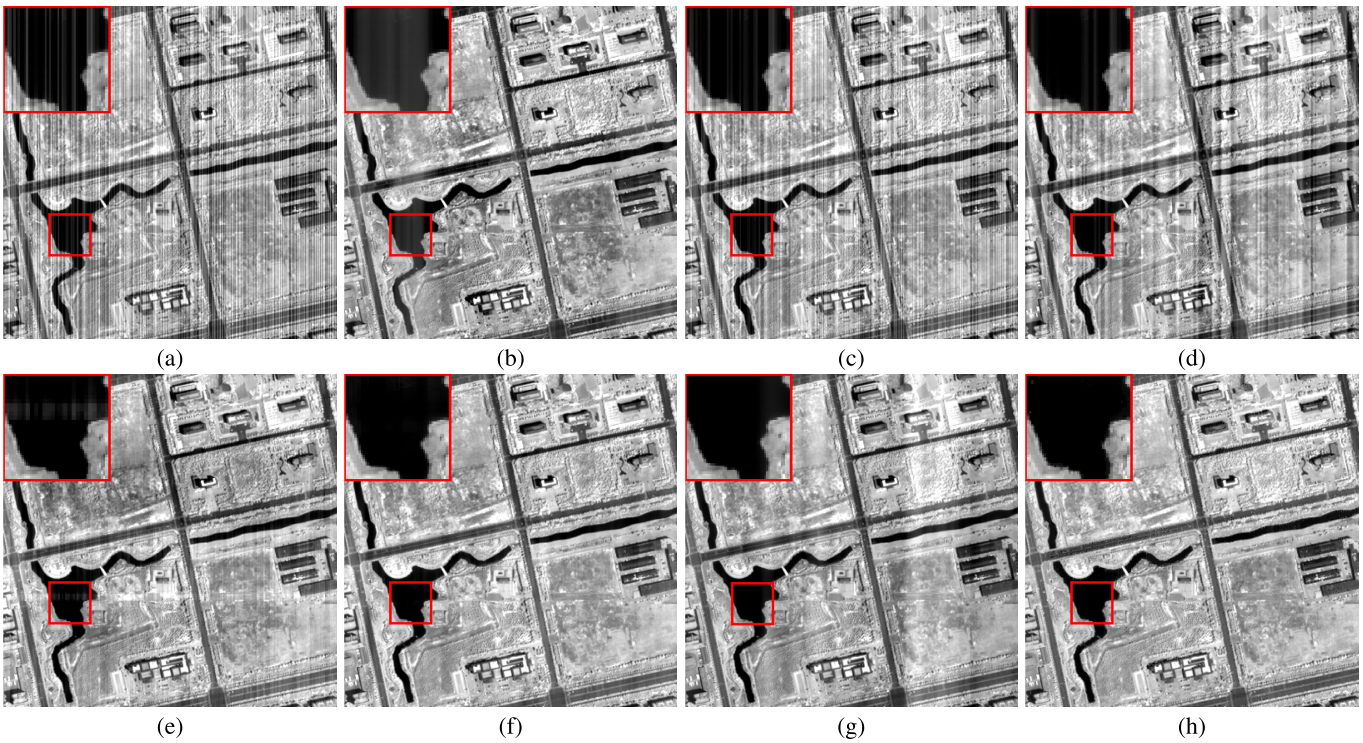


Fig. 10. Corrected Test2 for band 17. (a) Original noise image. (b) MM. (c) DIRN. (d) DnRCNN. (e) DLR. (f) ATVDLR. (g) $\ell_{2,1}$-SSTV. (h) Our method.

of the stripe noise removal capacity, effective information retention capacity, and operational efficiency of the proposed method.

Based on the stripe noise removal results of each algorithm (Figs. 9–12) and combined with the MNR, MPSNR, MSSIM, and MSAD results of each algorithm displayed in Table I, we focus on exploring the stripe noise removal capacity and effective information retention capacity of the proposed method when presented with different types of stripe noise.

We first analyzed the stripe noise removal and effective information retention capacities of DIRN, DnRCNN, and $\ell_{2,1}$-SSTV. From the denoising results of DIRN shown

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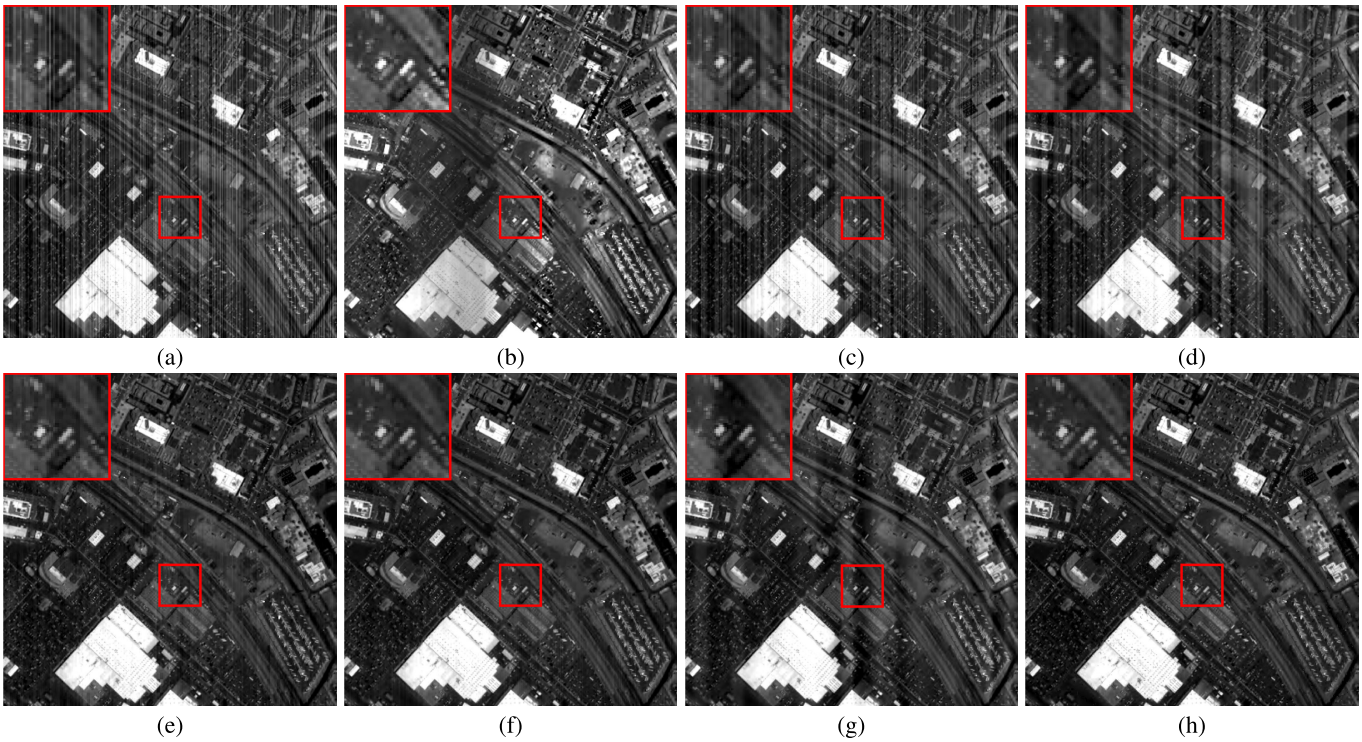


Fig. 11. Corrected Test3 for band 8. (a) Original noise image. (b) MM. (c) DIRN. (d) DnRCNN. (e) DLR. (f) ATVDLR. (g) $\ell_{2,1}$-SSTV. (h) Our method.

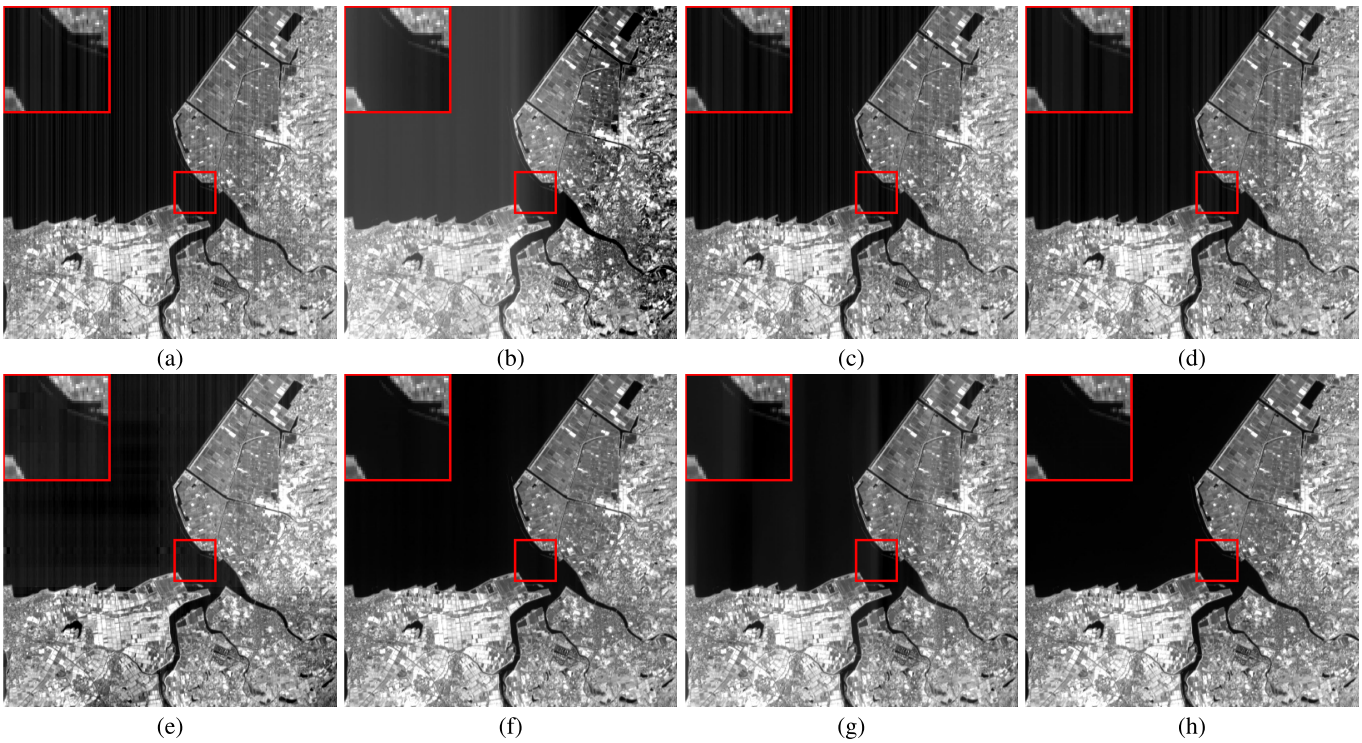


Fig. 12. Corrected Test4 for band 6. (a) Original noise image. (b) MM. (c) DIRN. (d) DnRCNN. (e) DLR. (f) ATVDLR. (g) $\ell_{2,1}$-SSTV. (h) Our method.

in Figs. 9(c)–12(c), the denoising results of DnRCNN in Figs. 9(d)–12(d), the denoising results of $\ell_{2,1}$-SSTV in Figs. 9(e)–12(e), and the MNR, MPSNR, MSSIM, and MSAD values of DIRN, DnRCNN, and $\ell_{2,1}$-SSTV listed in Table I, we can conclude that DIRN, DnRCNN, and $\ell_{2,1}$-SSTV do not produce satisfactory results for stripe noise removal from the four real HSIs shown in Fig. 8. Several reasons for this poor performance are possible. DIRN and DnRCNN, two deep learning-based methods, generate paired training data by adding simulated stripe noise of a certain intensity to clear images during training of the denoising network. However, multiple modes of stripe noise exist in real situations. Without

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TABLE I
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT DENOISING METHODS ON REAL HYPERSPECTRAL DATASETS

Image	Indicators	MM	DIRN	DnRCNN	DLR	ATVDLR	$\ell_{2,1}$-SSTV	Our
Test1	MNR	1.0934	1.1654	1.7049	1.7029	1.7168	1.7056	**1.7254**
	MPSNR	30.0394	32.2875	25.4885	30.9114	36.0001	28.8966	**36.9112**
	MSSIM	0.9218	0.9261	0.8536	0.8905	0.9358	0.9196	**0.9896**
	MSAD	10.8742	10.0068	13.5720	10.5475	6.4876	9.7392	**4.1453**
Test2	MNR	1.0773	1.2298	1.1914	1.6585	1.7169	1.7888	**1.8639**
	MPSNR	25.1359	25.4249	19.3407	25.8841	33.3373	27.8301	**34.1711**
	MSSIM	0.9198	0.9234	0.7192	0.8675	0.9346	0.9309	**0.9843**
	MSAD	12.3496	12.1692	16.4785	13.2555	8.4315	11.2754	**5.8562**
Test3	MNR	1.0876	1.2396	1.3243	1.4831	1.6500	1.5739	**1.6504**
	MPSNR	31.7735	31.8046	27.4363	29.2584	38.0707	35.1562	**38.8073**
	MSSIM	0.9455	0.9649	0.8413	0.9070	0.9872	0.9813	**0.9890**
	MSAD	9.3897	9.3675	12.1324	11.5394	9.1241	9.3371	**8.3879**
Test4	MNR	1.0418	1.0829	1.0278	1.1771	1.2989	1.1360	**1.3349**
	MPSNR	27.9980	28.1493	12.6219	27.8108	35.7372	22.4640	**39.8082**
	MSSIM	0.9396	0.9493	0.5538	0.9196	0.9912	0.8643	**0.9975**
	MSAD	11.5896	11.9170	35.1617	12.8383	8.9613	15.9752	**3.0426**

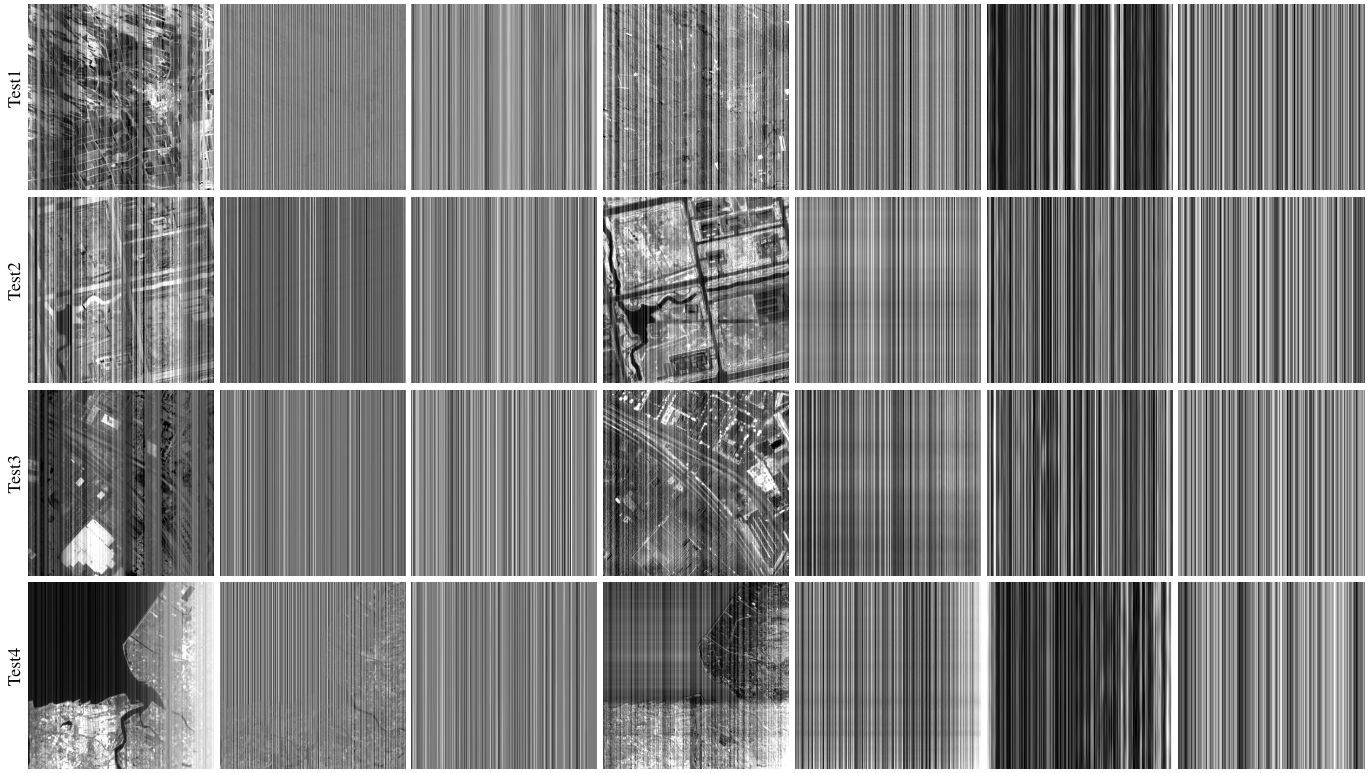


Fig. 13. Strip noise of four pairs of real hyperspectral data extracted by each method. (a) MM. (b) DIRN. (c) DnRCNN. (d) DLR. (e) ATVDLR. (f) $\ell_{2,1}$-SSTV. (g) Our method.

all possible stripe noise modes being fully considered or covered during the training process, the stripe noise cannot be effectively removed. Furthermore, $\ell_{2,1}$-SSTV is an HSI stripe noise removal method based on $\ell_{2,1}$ norm minimization and SSTV regularization. This method is sensitive to the selection of regularization parameters. Different parameter settings may lead to significant differences in results, and therefore further work is needed to select the optimal parameters. If parameter values are not selected properly, effective removal of stripe noise cannot be achieved.

Next, we analyzed the stripe noise removal and effective information retention capacities of MM and DLR. Based on the denoising results of each method and the data presented in Table I, MM and DLR consistently obtained favorable stripe noise removal results when processing real HSIs with relatively simple ground object types and small grayscale changes [Fig. 8(a)]. However, when processing real HSIs with relatively complex ground object types and large grayscale changes [Fig. 8(b)–(d)], those two methods did not achieve complete removal of stripe noise and caused varying degrees

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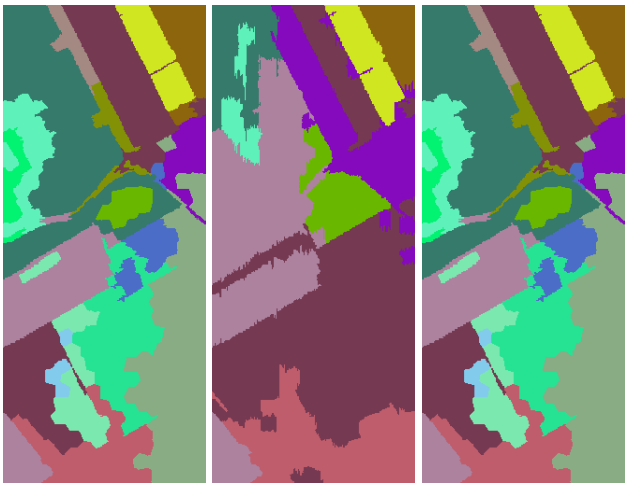
TABLE II
OPERATIONAL EFFICIENCY OF DIFFERENT DENOISING METHODS

Indicator/s	MM	DIRN	DnRCNN	DLR	ATVDLR	$\ell_{2,1}$-SSTV	Our
T	11.52	0.30	0.97	85.73	247.75	63.97	165.49

of damage to the effective information contained in the image. This discrepancy is mainly related to the limitations of MM and DLR. MM assumes that the digital distributions of the imaging system detectors are the same and uses differences in grayscale statistical information between the stripe noise-polluted area and the nonpolluted area to make the statistical information of the stripe noise-polluted area closer to that of the nonpolluted area, thereby achieving the removal of stripe noise. However, its assumptions are too restrictive for complex images and are suitable only for processing of images with relatively simple ground object types and small grayscale changes. When processing images with relatively complex ground object types and large grayscale changes, stripe artifacts are often produced, and the denoising effect is poor. DLR is an algorithm based on dual low-rank matrix decomposition. This algorithm uses the sparsity of stripe noise to decompose the HSI into the sum of multiple low-rank matrices, and then captures and removes stripe noise through sparse representation. However, this method has high computational complexity and a strong dependence on parameters. Different parameter settings must be used for different types of stripe noise, which limits its effectiveness in engineering applications. From the perspective of denoising results alone, the effects of the method proposed in this study and ATVDLR are best. Both of these methods achieved complete removal of multiple types of stripe noise and maintained the effective information of the image better than other methods. However, comparison and analysis of the indicators in Table I clearly shows that the results of MNR, MPSNR, MSSIM, and MSAD for the proposed method are optimal. This excellent performance is due to the method proposed in this study making full use of the spectral information of the HSI. After selection of the reference image, we use the proposed DMIV module and intensity adaptive grayscale value reconstruction algorithm to achieve the classification of stripe noise throughout the spectrum prior to processing, allowing the stripe noise in various complex scenes to be removed while maintaining the effective information of the image. In particular, when processing HSIs with large grayscale fluctuations, the proposed method has better stripe noise removal performance than other tested methods.

To more comprehensively evaluate the algorithms presented in this study, we compared and analyzed the operational efficiency (T) of each algorithm using real HSIs, as shown in Fig. 8. T is defined as the average time required for each method to process the four real HSIs shown in Fig. 8, and the results are presented in Table II.

Table II indicates that the processing speed of our method is not the fastest among methods, with DIRN and DnRCNN being notably faster. This difference arises mainly because the method in this study requires substantial time for pixel threshold classification and noise intensity estimation of multiple pixel classes. Although our method does not achieve the best operating efficiency, compared to other methods, it can effectively handle random stripe noise in various complex scenarios, particularly for HSIs with large grayscale fluctuations. Fig. 13 shows the stripe noise extraction results of each method for the real HSIs presented in Fig. 8, from which we can observe that stripe noise extracted via the method described in this study is complete and regular, while also containing negligible effective information of the image.



(a) (b) (c)

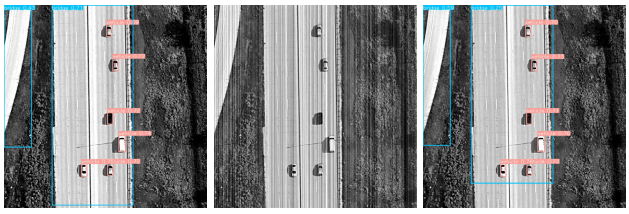
Fig. 14. Evaluation of destriping results of the proposed method on segmentation task. Segmentation results for (a) original image, (b) striped image, and (c) corrected image.

C. Effectiveness for High-Level Tasks

In addition to effectively improving the visual effect of remote sensing images, destriping can facilitate the subsequent application of remote sensing images to advanced tasks. Therefore, in this section, we further demonstrate the effectiveness of this method for high-level tasks, such as segmentation and object detection.

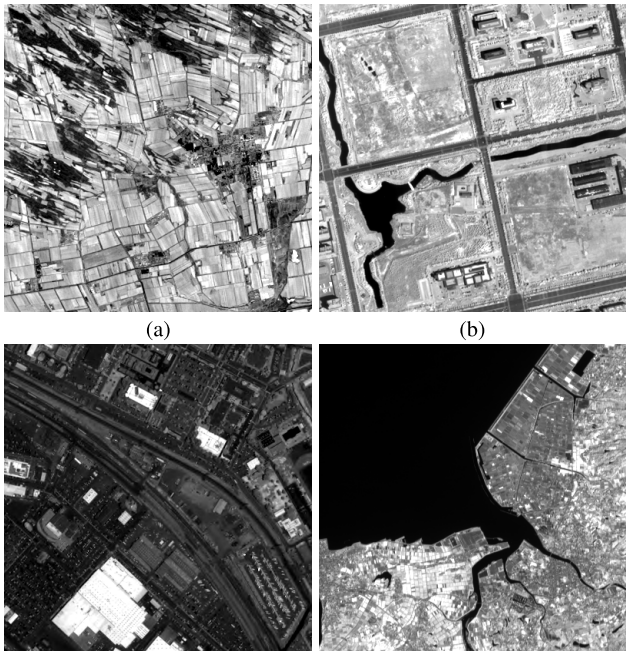
We use superpixel-based and spatially regularized diffusion learning (SSDL) [37] to perform semantic segmentation on the original image, stripe noise image, and stripe noise-free image of Salinas, and the results are presented in Fig. 14. As shown in Fig. 14(b), stripe noise seriously affects the results of semantic segmentation. This impact occurs because the presence of stripe noise damages the spectral information of various objects, resulting in errors in the clustering process. After removing stripe noise using the method proposed in this study, the stripe-free image was semantically segmented using SSDL. The segmentation results are shown in Fig. 14(c), which illustrates that the segmentation results are significantly improved and nearly consistent with the segmentation results of the original image, shown in Fig. 14(a). In terms of target detection, we use You Only Look Once version 8 (YOLO-V8) to detect targets in images from the remote sensing target detection dataset Detection in Optical Remote Sensing Images, and the results are shown in Fig. 15. This figure shows that stripe noise seriously damages the image features of cars and roads, which makes the detection algorithm being unable to identify vehicles and roads in noisy images. After removing

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(a) (b) (c)

Fig. 15. Evaluation of destriping results of the proposed method on object detection task. Detection results for (a) original image, (b) striped image, and (c) corrected image.



(a) (b) (c) (d)

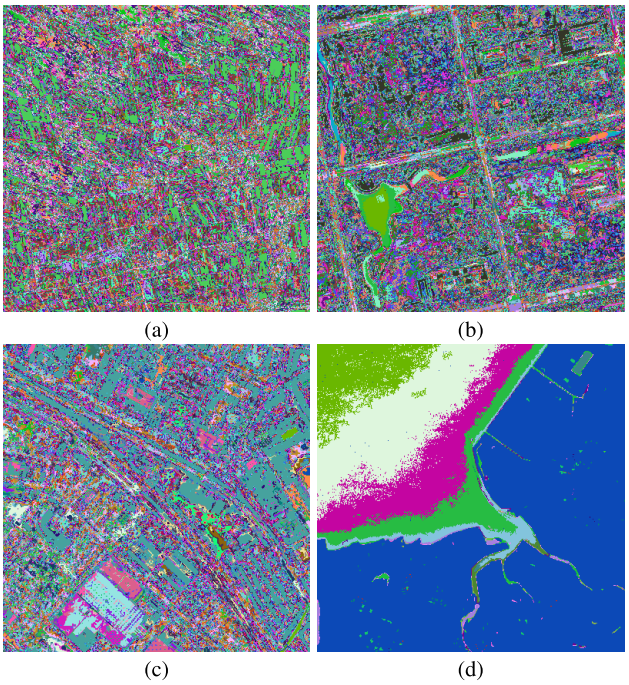
Fig. 16. Reference images. (a) Reference image for Test1. (b) Reference image for Test2. (c) Reference image for Test3. (d) Reference image for Test4.

stripe noise using the method described in this study, the detection algorithm effectively identified vehicles and roads in the image, and the recognition results were identical to those of the original image. Therefore, our method can effectively support the execution of subsequent high-level tasks.

D. Evaluation of DMIV Threshold Classification Accuracy

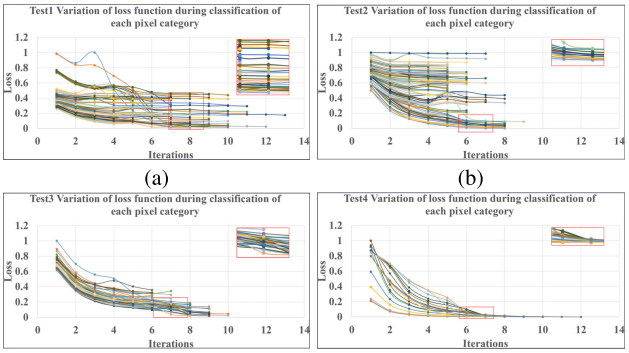
To further explore the effectiveness of the proposed method, we use the DMIV-based pixel threshold classification method to classify pixels of the four reference images shown in Fig. 16. The pixel threshold classification results for each reference image, along with the loss function for each pixel category threshold classification process, are thus obtained and analyzed in detail.

In Fig. 17, we show the classification results of the pixel threshold classification method based on DMIV. The proposed method obtains accurate threshold classification results when classifying pixels of HSIs in various complex scenes: Test1 is divided into 69 categories, Test2 is divided into 85 categories, Test3 is divided into 63 categories, and Test4 is divided into 21 categories. To better assess the performance of the proposed threshold classification method, we also determine changes in the loss function during the threshold classification process of



(a) (b) (c) (d)

Fig. 17. DMIV threshold classification results of (a) Test1, (b) Test2, (c) Test3, and (d) Test4.



(a) (b) (c) (d)

Fig. 18. Loss function for each pixel category in DMIV threshold classification process. (a) Test1. (b) Test2. (c) Test3. (d) Test4.

each pixel category, as shown in Fig. 18. As the number of iterations increases, the loss function of each pixel category gradually decreases and eventually stabilizes, reflecting the accuracy of threshold classification in the proposed method from another perspective.

After accurately classifying the pixels of the reference image, corresponding pseudo-labels are generated according to the threshold classification results and used to classify pixels in images contaminated by random stripe noise in adjacent bands. After threshold classification is completed, the intensity adaptive grayscale value reconstruction algorithm proposed in this study is used to remove stripe noise in the same pixel category, thereby achieving optimal denoising. This efficacy is also demonstrated by the denoising results of each method and the extracted stripe noise.

E. Performance Analysis of Standard and Improved MM Algorithms

Next, we selected two areas 400 × 400 pixels in size from the 16-bit HSI shown in Fig. 19, which contain uniform

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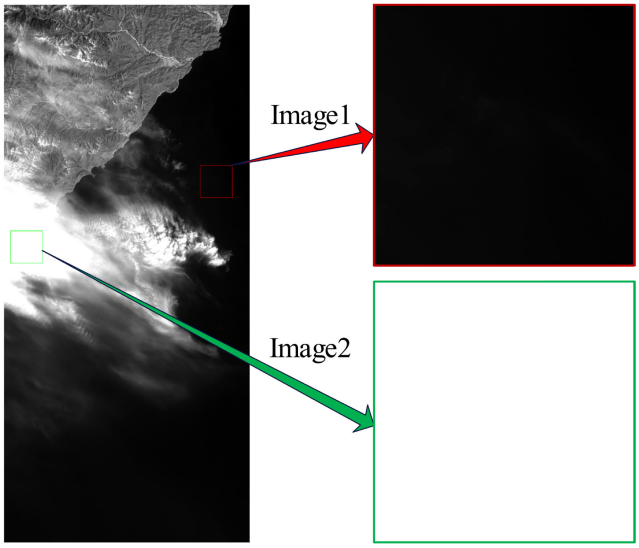


Fig. 19. Selected homogeneous region, Image1 and Image2.

TABLE III
EVALUATION OF DENOISING EFFECTS OF MM AND PFMM ON IMAGE1

Indicators	Image	MM	Mean_MM	PFMM
PSNR	Image1_1	90.0461	85.2878	**92.8057**
	Image1_2	89.6126	71.9171	**92.0185**
	Image1_3	85.3380	60.1907	**85.4937**
	Image1_4	**84.3085**	58.2423	70.7936
	Image1_5	**82.1659**	56.1932	67.3647
SSIM	Image1_1	0.9988	0.9774	**0.9990**
	Image1_2	0.9978	0.6463	**0.9983**
	Image1_3	0.9781	0.3795	**0.9836**
	Image1_4	**0.9847**	0.1886	0.9810
	Image1_5	**0.9748**	0.0627	0.9703

regions such as sea water (Image1) and thick clouds (Image2). By adding random stripe noise of various intensities to the selected areas, we explored the effects of the standard MM method [11], the improved MM method based on mean filtering (Mean_MM) [31], and the improved MM method based on polynomial fitting (PFMM) [32] on removal of stripe noise of various intensities.

The generated stripe noise images are recorded as Image1_1 ($r = 0.1, I = 0.1$), Image1_2 ($r = 0.3, I = 0.3$), Image1_3 ($r = 0.5, I = 0.5$), Image1_4 ($r = 0.7, I = 0.7$), Image1_5 ($r = 0.9, I = 0.9$), Image2_1 ($r = 0.2, I = 0.2$), Image2_2 ($r = 0.4, I = 0.4$), Image2_3 ($r = 0.6, I = 0.6$), Image2_4 ($r = 0.8, I = 0.8$), and Image2_5 ($r = 1.0, I = 1.0$), where r and I represent the percentage and intensity of added noise, respectively. The denoising performance of each MM method was evaluated using peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM), and the results are presented in Tables III and IV.

Tables III and IV suggest that for stripe noise of various intensities, the denoising effect of the improved MM method based on mean filtering is poorest. The MM and PFMM methods exhibit different denoising performances as noise intensity increases; when the noise intensity is high, MM performs better, whereas when the noise intensity is low, PFMM performs better. To explore the intensity of stripe noise, the

TABLE IV
EVALUATION OF DENOISING EFFECTS OF MM AND PFMM ON IMAGE2

Indicators	Image	MM	Mean_MM	PFMM
PSNR	Image2_1	78.4461	85.3936	**89.7384**
	Image2_2	78.4147	71.9255	**89.3321**
	Image2_3	77.8946	60.2304	**84.7361**
	Image2_4	**77.3008**	58.2702	70.7943
	Image2_5	**77.1253**	56.2053	67.3653
SSIM	Image2_1	0.9956	0.9802	**0.9974**
	Image2_2	0.9947	0.6780	**0.9966**
	Image2_3	0.9928	0.4137	**0.9954**
	Image2_4	**0.9875**	0.2095	0.9838
	Image2_5	**0.9834**	0.0738	0.9721

TABLE V
IMPACT OF EACH COMPONENT OF THE LOSS FUNCTION

L_{IQR}		✓	✓	✓	✓	✓	✓
L_{rang}	✓		✓	✓	✓	✓	✓
L_{MAD}	✓	✓		✓	✓	✓	✓
L_{std}	✓	✓	✓		✓	✓	✓
L_{HBR}	✓	✓	✓	✓		✓	✓
L_{CV}	✓	✓	✓	✓	✓		✓
MPSNR	30.9867	31.3698	31.3546	30.0416	31.7161	30.9495	**37.4245**

MM and PFMM methods are suitable for removing, we added stripe noise of ($r = 0.1, I = 0.1$) to ($r = 0.9, I = 0.9$) to Image1 and Image2. Then we used both methods to denoise the generated noisy images, calculated the PSNR and SSIM values of the denoised images, and ultimately plotted the line graph shown in Fig. 20. Clearly, when the noise intensity is high, the MM method is preferable, while when the noise intensity is low, PFMM is preferable.

V. ABLATION STUDIES

To obtain high-precision pixel classification results, we proposed a loss function to determine whether the optimal processing result is obtained. This function consists of six components: quartile difference, range, mean difference, standard deviation, outlier ratio, and coefficient of variation. To demonstrate the effectiveness of each component, we conducted six sets of ablation experiments. The experimental data are the four real hyperspectral noise images shown in Fig. 8. The quantitative indicator is the average value of MPSNR across the four images. The results are presented in Table V. The optimal MPSNR value is obtained without deleting any component of the loss function; when any part of the loss function is deleted, MPSNR decreases, indicating that the denoising performance of the proposed method decreases when any component is deleted. This finding supports the importance of each component of the proposed loss function.

In addition, in order to more intuitively demonstrate the effectiveness of each component of the loss function in this article, we also give the DMIV pixel threshold classification results of four reference images (as shown in Fig. 16) when each component of the loss function is deleted separately, as shown in Fig. 21. From the pixel threshold classification results of DMIV in Fig. 21, it can be seen that when any component of the loss function is deleted, the number of pixel

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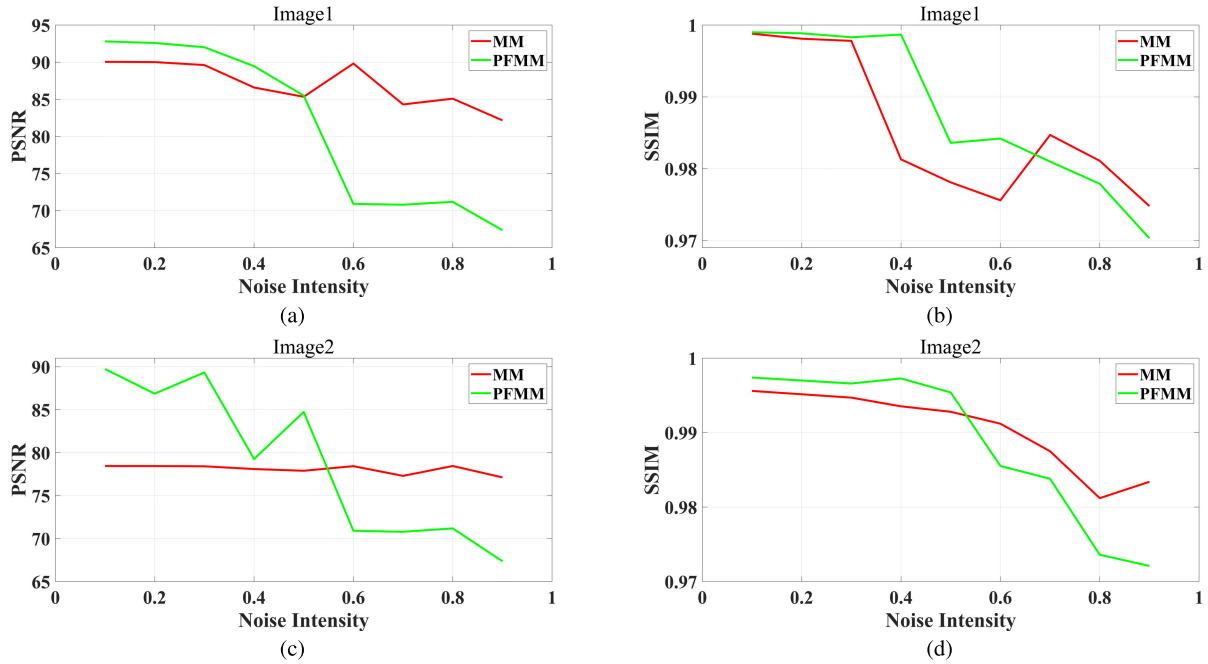


Fig. 20. Comparison of denoising effects between MM and PFMM. (a) and (b) Evaluation of denoising effects of MM and PFMM on Image1 as noise intensity increases. (c) and (d) Evaluation of denoising effects of MM and PFMM on Image2 as noise intensity increases.

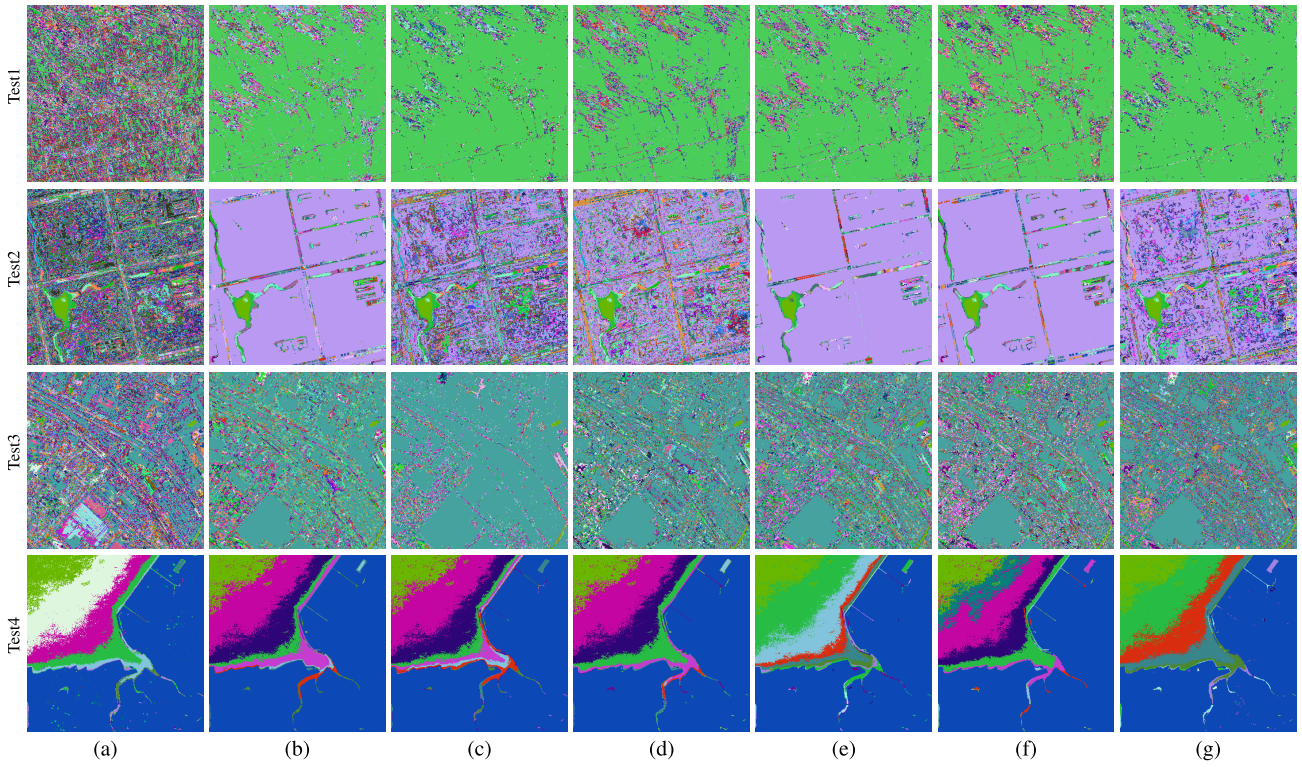


Fig. 21. Pixel threshold classification results of DMIV for the reference images (as shown in Fig. 16) of four real hyperspectral noise images when each component of the loss function is deleted separately. (a) Pixel threshold classification results when no component of the loss function is deleted. (b) L_{IQR}. (c) L_{range}. (d) L_{MAD}. (e) L_{std}. (f) L_{HBR}. (g) L_{CV}.

categories in the pixel threshold classification results of DMIV is significantly reduced. At the same time, we also give the denoising results of four real hyperspectral noise images when each component of the loss function is deleted separately, as shown in Fig. 22. From this, we can clearly see that after deleting any component of the loss function, the denoising results of the noise image are adversely affected, causing different degrees of grayscale distortion in the denoised image, which is caused by the insufficient accuracy of the pixel threshold classification. Therefore, when any component of the loss function in this article is removed, the denoising effect of the algorithm in this article will be affected. This intuitively

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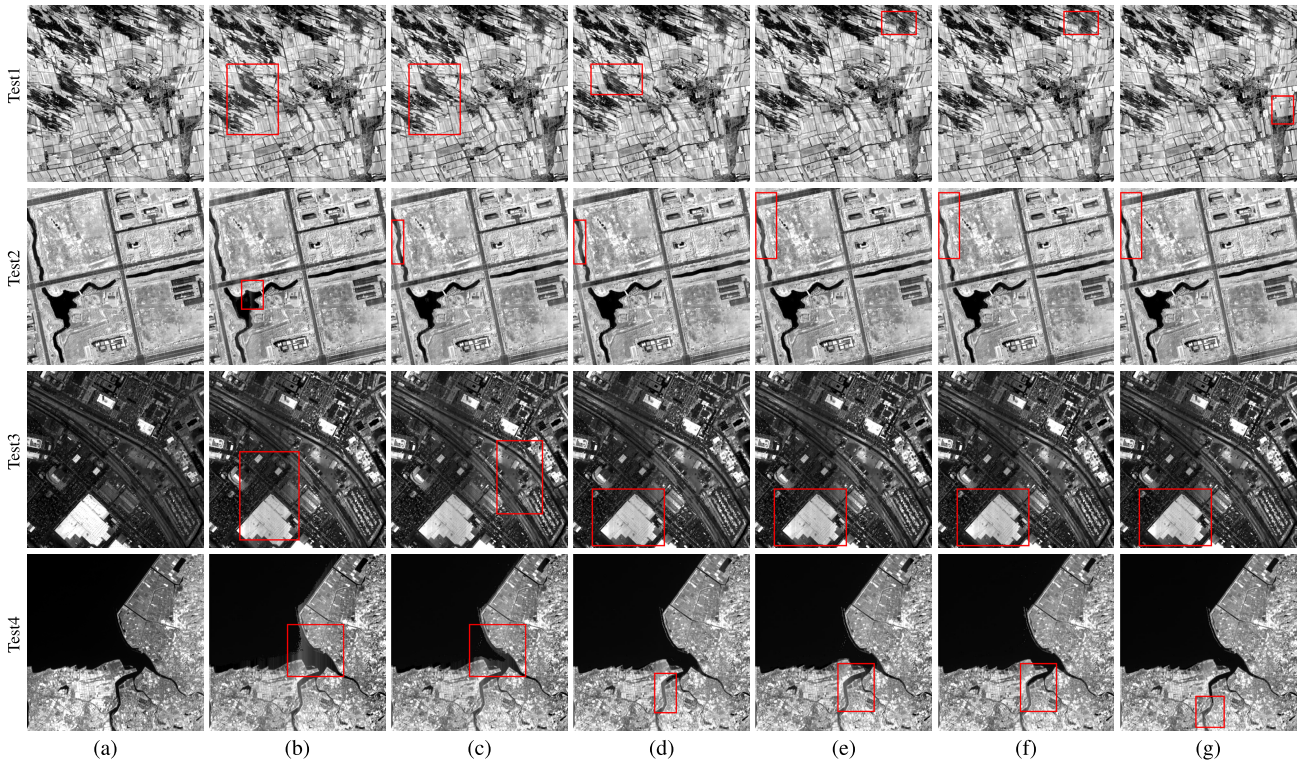


Fig. 22. Denoising results of four real hyperspectral noise images when each component of the loss function is deleted separately. (a) Denoising images when no component of the loss function is deleted. (b) L_{IQR}. (c) L_{range}. (d) L_{MAD}. (e) L_{std}. (f) L_{HBR}. (g) L_{CV}.

shows the effectiveness of each component of the loss function from another perspective.

VI. Conclusion

We propose an HSI intensity adaptive destriping method based on reference image-guided pixel clustering. We first analyze the HSI, manually select the initial reference image, and locate its specific position. Then, a pixel threshold classification method based on DMIV is used to classify the pixels of the selected reference image and corresponding pseudo-labels are generated according to the threshold classification results, which are used to guide the pixel threshold classification of a certain band that is contaminated by random stripe noise adjacent to the reference image. Finally, the resulting pixel categories are divided into high-intensity and low-intensity classes based on the noise intensity. The proposed intensity adaptive grayscale value reconstruction algorithm is used to remove stripe noise from each pixel category, and the denoised image is utilized as a new reference image. Starting from the initial reference image, these steps are repeated along the spectrum to achieve the removal of stripe noise from all bands. The performance of the proposed method is compared and analyzed against six state-of-the-art stripe removal algorithms using a real hyperspectral dataset. The experimental results show that the proposed method can completely remove stripe noise while protecting the effective information of the HSI. In particular, this method is useful for HSIs with large grayscale fluctuations, where it demonstrates superior stripe removal performance to established methods. As the method requires manual selection of appropriate initial reference images before denoising, the degree of automation possible is limited. We will conduct in-depth research on the automatic selection of initial reference images in the future.

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